



Tutorials

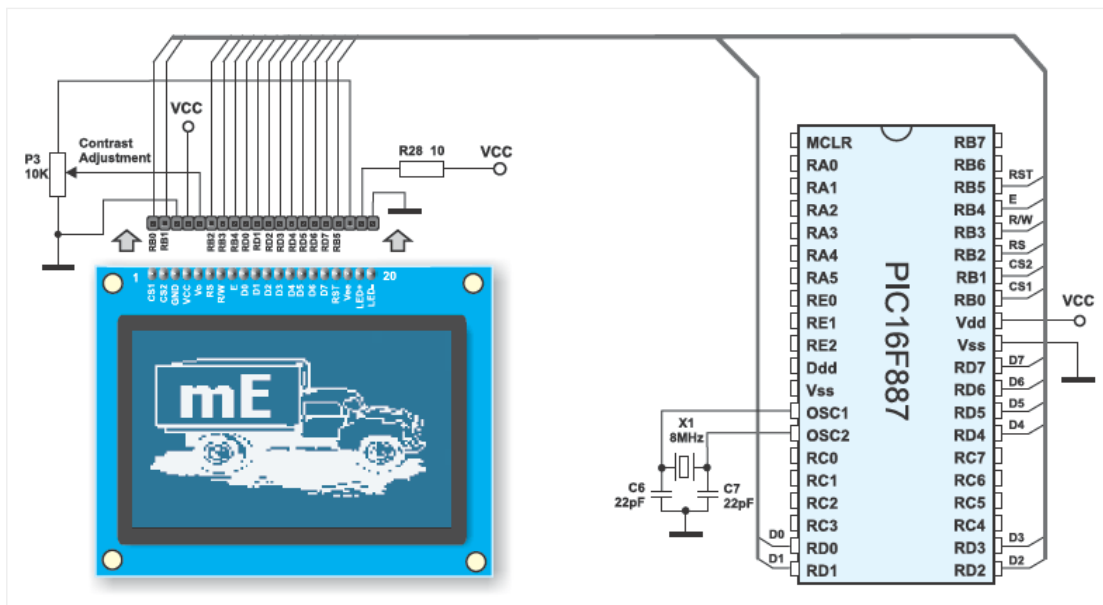
## 4.16 EXAMPLE 14

### MIKROELEKTRONIKA

#### Using graphic LCD display

A graphic LCD (GLCD) provides an advanced method for displaying visual messages. While the character LCD can display only alphanumeric characters, the GLCD can display messages in the form of drawings and bitmaps. The most commonly used graphic LCD has 128×64 pixels screen resolution. The GLCD contrast can be adjusted using the potentiometer P1.

Here, the GLCD displays a picture of truck the bitmap of which is stored in the `truck_bmp.c` file.



```

1  /*Header*****//
2
3  //Declarations-----
4  const code char truck_bmp[1024];
5  //-----end-declarations
6
7  // Glcd module connections
8  char GLCD_DataPort at PORTD;
9  sbit GLCD_CS1 at RB0_bit;
10 sbit GLCD_CS2 at RB1_bit;
11 sbit GLCD_RS at RB2_bit;
12 sbit GLCD_RW at RB3_bit;
13 sbit GLCD_EN at RB4_bit;
14 sbit GLCD_RST at RB5_bit;
15 sbit GLCD_CS1_Direction at TRISB0_bit;
16 sbit GLCD_CS2_Direction at TRISB1_bit;
17 sbit GLCD_RS_Direction at TRISB2_bit;
18 sbit GLCD_RW_Direction at TRISB3_bit;
19 sbit GLCD_EN_Direction at TRISB4_bit;
20 sbit GLCD_RST_Direction at TRISB5_bit;
21 // End Glcd module connections
22
23 void delay2S(){ // 2 second delay function
24     Delay_ms(2000);
25 }
26

```

```

27 void main() {
28     unsigned short ii;
29     char *someText;
30
31     #define COMPLETE_EXAMPLE
32     ANSEL = 0; // Configure AN pins as digital
33     ANSELH = 0;
34     C1ON_bit = 0; // Disable comparators
35     C2ON_bit = 0;
36     Glcd_Init(); // Initialize GLCD
37     Glcd_Fill(0x00); // Clear GLCD
38
39     while(1) {
40         #ifdef COMPLETE_EXAMPLE
41             Glcd_Image(truck_bmp); // Draw image
42             delay2S(); delay2S();
43         #endif
44
45         Glcd_Fill(0x00); // Clear GLCD
46         Glcd_Box(62,40,124,56,1); // Draw box
47         Glcd_Rectangle(5,5,84,35,1); // Draw rectangle
48         Glcd_Line(0, 0, 127, 63, 1); // Draw line
49         delay2S();
50
51         for(ii = 5; ii <= 60; ii+=5){ // Draw horizontal and vertical lines
52             Delay_ms(250);
53             Glcd_V_Line(2, 54, ii, 1);
54             Glcd_H_Line(2, 120, ii, 1);
55         }
56         delay2S();
57
58         Glcd_Fill(0x00); // Clear GLCD
59
60         #ifdef COMPLETE_EXAMPLE
61             Glcd_Set_Font(Character8x7, 8, 7, 32); // Choose font, see __Lib_GLCDFonts.c
62             // in Uses folder
63         #endif
64
65         Glcd_Write_Text("mikroE", 1, 7, 2); // Write string
66
67         for (ii = 1; ii <= 10; ii++) // Draw circles
68             Glcd_Circle(63,32, 3*ii, 1);
69         delay2S();
70
71         Glcd_Box(12,20, 70,57, 2); // Draw box
72         delay2S();
73
74         #ifdef COMPLETE_EXAMPLE
75             Glcd_Fill(0xFF); // Fill GLCD
76             Glcd_Set_Font(Character8x7, 8, 7, 32); // Change font
77             someText = "8x7 Font";
78             Glcd_Write_Text(someText, 5, 0, 2); // Write string
79             delay2S();
80
81             Glcd_Set_Font(System3x5, 3, 5, 32); // Change font
82             someText = "3X5 CAPITALS ONLY";
83             Glcd_Write_Text(someText, 60, 2, 2); // Write string
84             delay2S();
85
86             Glcd_Set_Font(font5x7, 5, 7, 32); // Change font
87             someText = "5x7 Font";
88             Glcd_Write_Text(someText, 5, 4, 2); // Write string
89             delay2S();
90
91             Glcd_Set_Font(FontSystem5x7_v2, 5, 7, 32); // Change font
92             someText = "5x7 Font (v2)";
93             Glcd_Write_Text(someText, 5, 6, 2); // Write string
94             delay2S();
95         #endif
96     }
97 }

```

truck\_bmp.c file:

```

1 // -----
2 // GLCD Picture name: truck.bmp
3 // GLCD Model: KS0108 128x64
4 // -----
5
6 unsigned char const truck_bmp[1024] = {
7     0, 0, 0, 0, 0, 248, 8, 8, 8, 8, 8, 8, 12, 12, 12, 12,
8     12, 10, 10, 10, 10, 10, 10, 9, 9, 9, 9, 9, 9, 9, 9, 9,
9     9, 9, 9, 9, 9, 9, 9, 9, 9, 137, 137, 137, 137, 137, 137,
10    137, 137, 137, 137, 137, 137, 137, 9, 9, 9, 9, 9, 9, 9, 9,
11    9, 9, 13, 253, 13, 195, 6, 252, 0, 0, 0, 0, 0, 0, 0, 0,
12    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
13    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
14    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
15    0, 0, 0, 0, 0, 255, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
16    240, 240, 240, 240, 240, 224, 224, 240, 240, 240, 240, 224, 192, 192, 224,
17    240, 240, 240, 240, 240, 224, 192, 0, 0, 0, 255, 255, 255, 255, 255, 195,
18    195, 195, 195, 195, 195, 195, 3, 0, 0, 0, 0, 0, 0, 0, 0,
19    0, 0, 0, 255, 240, 79, 224, 255, 96, 96, 96, 32, 32, 32, 32,
20    32, 32, 32, 32, 32, 32, 32, 32, 64, 64, 64, 64, 128, 0, 0, 0,
21    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
22    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
23    0, 0, 0, 0, 0, 255, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
24    255, 255, 255, 255, 255, 0, 0, 0, 0, 255, 255, 255, 255, 255, 0, 0,

```

```

25 0, 0,255,255,255,255,255, 0, 0, 0,255,255,255,255,255,129,
26 129,129,129,129,129,129,128, 0, 0, 0, 0, 0, 0, 0, 0,
27 0, 0, 0,255, 1,248, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8,
28 8, 8, 8, 8, 16,224, 24, 36,196, 70,130,130,133,217,102,112,
29 160,192, 96, 96, 32, 32,160,160,224,224,192, 64, 64,128,128,192,
30 64,128, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
31 0, 0, 0, 0, 0, 63, 96, 96, 96,224, 96, 96, 96, 96, 96,
32 99, 99, 99, 99, 99, 96, 96, 96, 96, 99, 99, 99, 99, 96, 96,
33 96, 96, 99, 99, 99, 99, 96, 96, 96, 96, 99, 99, 99, 99, 99,
34 99, 99, 99, 99, 99, 99, 96, 96, 96, 96, 96, 96, 64, 64,
35 64,224,224,255,246, 1, 14, 6, 6, 2, 2, 2, 2, 2, 2,
36 2, 2, 2, 2,130, 67,114, 62, 35, 16, 16, 0, 7, 3, 3, 2,
37 4, 4, 4, 4, 4, 4, 28, 16, 16, 16, 17, 17, 9, 9, 41,
38 112, 32, 67, 5,240,126,174,128, 56, 0, 0, 0, 0, 0, 0,
39 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1,
40 1, 1,127,127,127,127,255,255,247,251,123,191, 95, 93,125,189,
41 189, 63, 93, 89,177,115,243,229,207, 27, 63,119,255,207,191,255,
42 255,255,255,255,255,255,255,255,127,127,127,127,127,127,255,
43 255,255,127,127,125,120,120,120,120,120,248,120,120,120,120,
44 120,248,248,232,143, 0, 0, 0, 0, 0, 0, 0,128,240,248,
45 120,188,220, 92,252, 28, 28, 60, 92, 60,120,248,248, 96,192,
46 143,168,216,136, 49, 68, 72, 50,160, 96, 0, 0, 0, 0, 0,
47 0, 0, 0,128,192,248,248,248,248,252,254,254,254,254,254,
48 254,254,254,254,254,255,255,255,255,255,246,239,208,246,174,173,
49 169,128,209,208,224,247,249,255,255,252,220,240,127,255,223,255,
50 255,255,255,255,255,254,254,255,255,255,255,255,255,254,255,
51 255,255,255,255,255,255,254,254,254,254,254,254,254,254,254,
52 254,254,254,254,255,255,255,255,255,254,255,190,255,255,253,
53 240,239,221,223,254,168,136,170,196,208,228,230,248,127,126,156,
54 223,226,242,242,242,242,177, 32, 0, 0, 0, 0, 0, 0,
55 0, 0, 0, 1, 1, 1, 1, 3, 3, 3, 7, 7, 7, 7, 15,
56 15, 15, 7, 15, 15, 15, 7, 7, 15, 14, 15, 13, 15, 47, 43, 43,
57 43, 43, 43, 47,111,239,255,253,253,255,254,255,255,255,255,
58 191,191,239,239,239,191,255,191,255,255,255,255,255,255,255,
59 255,255,255,255,255,255,255,255,255,255,255,255,255,255,
60 255,255,255,255,127,127,127,127,255,255,191,191,191,255,254,
61 255,253,255,255,255,251,255,255,255,127,125, 63, 31, 31, 31, 31,
62 31, 31, 63, 15, 15, 7, 7, 3, 3, 3, 0, 0, 0, 0, 0, 0,
63 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
64 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
65 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0,
66 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1,
67 1, 1, 1, 1, 3, 3, 3, 11, 11, 11, 11, 7, 3, 14, 6, 6,
68 6, 2, 18, 19, 19, 3, 23, 21, 21, 17, 1, 19, 19, 3, 6, 6,
69 14, 15, 15, 7, 15, 15, 15, 11, 2, 0, 0, 0, 0, 0, 0, 0,
70 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0
71 };

```

In order to make this example work properly, it is necessary to tick off the GLCD library in the *Library Manager* prior to compiling. Also, it is necessary to include document *truck\_bmp.c* into the project.



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